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**ICT - 101**

## Module 3: Unit 3

1. A device manager serves as a central hub for managing and visualizing all installed hardware components on a computer. Users can view a list of all installed devices, together with information about their type, manufacturer, and status, as well as enable or disable them, update, or reinstall their drivers, troubleshoot device issues, and uninstall any extra devices.

Both inexperienced and experienced computer users will find the device manager to be a useful tool. It can be used to identify and resolve hardware issues, enhance performance, and maintain equipment current.

2. Your ability to pinpoint the precise actions you must take to check your network connection and configuration settings will depend on the sort of device you are using and its operating system.

Wi-Fi, a cellular network, or a wired network? You can troubleshoot any connection issues by knowing the network type and its network name. The name of the network you are connected to can be useful for troubleshooting.

3. False; you must use the Display settings to adjust the DPI of a display device. The Personalization program allows users to customize desktop themes and appearance.

4. Impact printers apply physical force to the paper to imprint an image. This can be accomplished by striking an inked ribbon with a hammer or by striking the paper directly with pins. Physical impact is not used by non-impact printers to imprint an image on the paper. Instead, they apply a number of techniques, including heat, inkjets, and lasers, to transfer the ink or toner to the paper. Heat is used by thermal printers to imprint an image on the paper. This can be accomplished by burning tiny dots into the paper or melting a wax-based coating that is applied to it. Plotters are specialty printers made for printing large-format documents like architectural plans and technical drawings. Plotters utilize a range of tools, such as pens, pencils, and markers, to draw an image on the paper.

5. These settings can be changed in the printer driver. A component of software called the printer driver gives you control over the printer. Right-click on the printer icon in the Devices and Printers pane and choose Printer settings to launch the printer driver. Click the Printing Preferences tab after launching the printer driver. Here, you can adjust the print parameters for quality and page order.

**Module 3: Unit 4**

1. System memory, commonly referred to as primary memory, is the short-term data storage used by a computer's CPU. Because it is volatile, all information in system memory is lost when the machine shuts down. Although system memory is incredibly quick, it is also relatively expensive and has a little amount of space.

Data that is not actively being used by the CPU is permanently stored in mass storage, commonly referred to as secondary storage. Data is kept even when the computer is shut off since it is non-volatile. While mass storage is more expensive and has a bigger capacity, it is slower than system memory.

2. System memory most often used is random-access memory (RAM). It is volatile, which means that when the computer is powered down, the data in RAM is lost. Because RAM is so speedy, the CPU can access data with ease and speed. The operating system, applications, and data that the computer is now using are kept in RAM.

Read-only memory (ROM): Since ROM is non-volatile, the data stored inside is kept even when the computer is powered down. Important system files like the BIOS and firmware are kept in ROM. Additionally, information that must always be available is stored in ROM, such as the BIOS's language preferences. In addition, unlike ROM memory, which cannot be overwritten, RAM memory allows for unrestricted, limitless value changes.

3. An SSD is the ideal option if you want the fastest and most reliable storage device. But the price of SSDs is also higher than that of HDDs. An HDD is a more cost-effective choice if you require a lot of storage space but are on a tight budget. The computer may use a portion of the hard drive for storage if there is not enough space to store all of the programs and data files that are needed. If the disk is slow, performance will suffer when the computer tries to load programs and data files into memory (read operations) or write data from memory to files system RAM (virtual memory) augmentation.

4. External hard disk drives (HDDs): HDDs are common storage gadgets that keep data on spinning platters. Solid-state external drives (SSDs): SSDs are more recent storage devices that store data on flash memory chips. Optical discs: Large amounts of data can be stored on optical discs like CDs, DVDs, and Blu-rays. Flash memory drives: Small, portable storage devices that can be used to store files and data include USB flash drives and SD cards. You can store files and data on distant servers using cloud storage, a sort of internet storage. Anywhere with an internet connection can access cloud storage, which is incredibly practical.

**Module 3: Unit 5**

1. **Compression** - the file system has the ability to automatically minimize disk space the area occupied by a file. Application of a non-lossy algorithm by the file system the file to identify ways to store the data more efficiently within it without eliminating whatever details.
**Encryption** - Data in a file may be automatically encrypted by the file system when it is saved. This signifies that access to the encryption key is required in order to open the file.
**Permissions** - Each file or folder object is kept up to date using an Access Control List (ACL) by the file system. Which user accounts are permitted to view, write, or control the item are listed in the ACL.
**Journaling** - The file system keeps a log of all intended and unintended changes. This means that if there is an unexpected power outage and a specific write activity was halted, the journal could be utilized to recover the data or at the very least return the file system to a stable condition.
**Limitations** - File systems have limitations in terms of their maximum capacity and the size of individual files, as shown in the table below.
**Naming rule** - The maximum file name length for extremely old file systems was eight characters plus a three-character extension.

2. FAT has a maximum field size of 4 GB, limited by partition size, limited to 255 users and for security it has Basic access control lists (ACLs), supports a variety of file types, has faster on small files and drives but has less reliable and its compatible with a wide range of operating systems. While NTFS has 16 exabytes of file size (16 billion GB), it has also a same maximum of files as FAT but has unlimited users and for its security it has Advanced ACLs, encryption, and auditing, for its features, performance, reliability and compatibility it has Supports a wider range of file types, including symbolic links and hard links, Faster on large files and drives, More reliable and Compatible with Windows and some other operating systems.

3.

a. .txt – Text File
b. .doc(x) – Microsoft Word Document
c. .pdf - Portable Document Format (PDF) file
d. .jpeg – Joint Photographic Experts Group (JPEG) image file
e. .png – Portable Network Graphics (PNG) image file
f. .gif – Graphics Interchange Format (GIF) image file
g. .mp3 – MPEG-1 Audio Layer 3 (MP3) audio file
h. .mp4 – MPEG-4 Part 14 (MP4) video file
j. .avi – Audio Videp Interleave (AVI) video file
i. .exe – Executable file
k. .zip – ZIP archive file

4. I will Identify the user's specific needs, and I will probably use windows features like file permission to configure it, and Encryption so that it cannot be read by unauthorized users.. A

password manager can help you create and manage strong passwords for all of your online accounts.